SUPERIOR UNIVERSITY

HEART DISEASE PREDICTION SYSTEM

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SUBJECT: Data Mining

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1. Introduction

Heart disease is one of the leading causes of death worldwide. Early prediction and diagnosis can help reduce risks and improve patient outcomes.

This project focuses on predicting heart disease using **Data Mining techniques**. It involves extracting meaningful patterns from medical datasets using machine learning algorithms. The system analyzes patient data such as age, cholesterol level, and blood pressure to predict whether a person is at risk of heart disease.

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2. Objectives

• Apply data mining techniques to analyze healthcare data
• Build predictive models using machine learning algorithms
• Identify patterns related to heart disease risk
• Evaluate model performance using accuracy and other metrics
• Develop a user-friendly web-based prediction system

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3. Features

• Interactive web interface for user input
• Real-time heart disease prediction
• Machine learning-based classification model
• Data preprocessing and feature handling
• Display of prediction results (High Risk / Low Risk)

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4. Tools and Technologies

• Programming Language: Python
• Web Framework: Flask
• Machine Learning: Scikit-learn
• Data Visualization: Matplotlib, Seaborn
• Data Handling: Pandas, NumPy
• Model Storage: Joblib

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5. Libraries Used

Library	Purpose
Pandas	Data manipulation and analysis
NumPy	Numerical computations
Scikit-learn	Machine learning models
Matplotlib	Data visualization
Seaborn	Advanced visualization
Flask	Web application framework
Joblib	Model saving and loading

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6. Data Mining Techniques Used

The following data mining techniques are applied in this project:

• **Classification:** Used to predict whether a patient has heart disease or not
• **Data Preprocessing:** Cleaning and preparing data for analysis
• **Feature Selection:** Identifying important attributes affecting prediction
• **Pattern Recognition:** Finding relationships between different medical features

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7. Methodology

The system follows these steps:

1. **Data Collection**
   - Heart disease dataset is used
2. **Data Preprocessing**
   - Handling missing values
   - Data normalization
   - Converting categorical values
3. **Feature Selection**
   - Selecting important features like age, cholesterol, etc.

4. **Model Training**
   - Applying machine learning algorithms such as Logistic Regression and Decision Tree
5. **Model Evaluation**
   - Measuring performance using accuracy, precision, recall
6. **Prediction System**
   - Flask web app takes user input and predicts results

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8. Data Analysis

The dataset includes important features such as:

Feature	Type	Importance
Age	Numerical	High
Sex	Categorical	Medium
Cholesterol	Numerical	High
Blood Pressure	Numerical	Medium
Max Heart Rate	Numerical	High

Analysis includes:

• Identifying relationships between features
• Observing patterns in high-risk patients
• Understanding how different attributes affect prediction

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9. System Workflow

Workflow Steps:

User Input → Data Preprocessing → Machine Learning Model → Prediction Output

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10. Results

The performance of the model is evaluated using the following metrics:

Metric	Value
Accuracy	85%
Precision	82%
Recall	88%
F1-Score	85%

The model shows good performance in predicting heart disease risk.

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11. Conclusion

This project demonstrates the use of **Data Mining techniques** in the healthcare domain. Machine learning models successfully predict heart disease risk based on patient data.

The system can help in early diagnosis and assist healthcare professionals in decision-making. Future improvements may include using advanced models and larger datasets for better accuracy.